Innovative design project

Problem when multiple evs are connected, the charging station can create a large and varying load . this can cause overloading , voltage fluctuations, inefficient power utilization and increased peak demand.

Develop a smart EV charge station that continuously monitors the available electrical power and ev charging demand , then dynamically distributs the available power among connected evs

Controller can consider

The grid /load power availability
No of connected evs
Ev battery soc
Charging priority
Requested charging power
Maximum station capacity
Peak load condition

Other version

Smart ev charging station with dynamic load balancing and priority based charging

Core research: how to distribute a limited available grid power among multiple evs in real time without overloading the feeder/charging station, while still satisfying ev users charging requirement

Recent research review: identify peak demand , feeder congestion, voltage issuses, thermal stress and coordinated/smart charging as important ev grid problem

This could be solved by

Peak demand---shift peak demand ---delays charging until off-peak hours

Relives feeder congestion and thermal stress: throttle down charging rated dynamically when local current approaches maximum rating limits

Fixes voltage issues: Balance load across three power phases to prevent deep voltage at remote areas

Evse: electric vehicle supply equipment

This is the charging station that supplies electrical energy to ev and handles the required control/safety functions

IEC 61851-1 is the one of the important

How does evse works

Ac----ac/dc rectifier-----dc/dc converter-----battery pack again back (bidirectional current flow)

Common evse level

Level1---single phase AC---120v/230v standard wall socket---1.4kw to 2.3kw (8-12 km range/hr)---home overnight trickle charging

Level2---single/three phase ac---230v/400v dedicated evse unit---3.7kw to 22 kw(25-100 kn range/hr)---commerical parking

Level3---direct current (DC)---400v to 800v---50kw to 350+ kw (80% charge 15-30 min)---highway charging hubs.

Evse board is the device running your control algorithm: moitirs building load---adjusts control pilots---executes dynamic commands

Control pilot pin is a communication signal line in AC charging connectors governed by IEC 61851-1 standard. Ir allows evse to continuously tell the ev the max allowed current limit using pulse width modulation

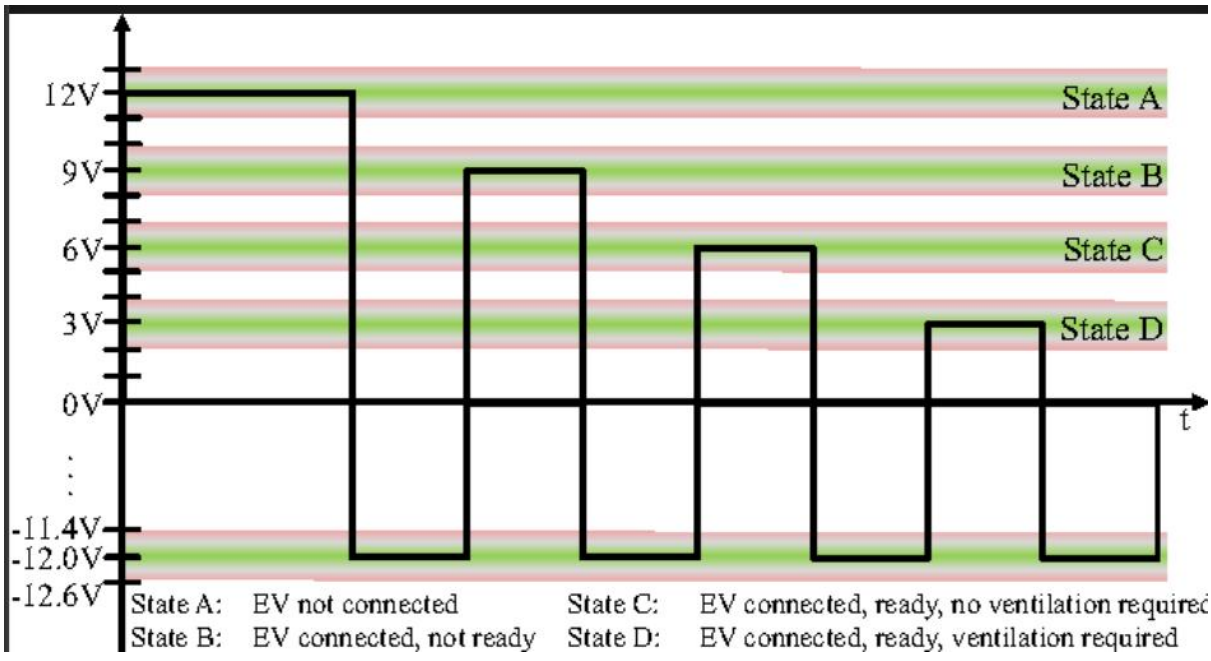
Step1: handshake and voltage states: before ac signal send , evse and ev communicate by changing dc voltage level on control pilot line using internal resistance

State A---+12 v (steady dc)---vechicle is not connected—mains power relay stays off

State B---+9v (DC or 1khz pwm)----plug inserted , ev detedcted---evse turns on the 1khz pwm signal

State c----+6v(1khz pwm)---vechicle ready to charge(ev request power)---relay to pass ac power

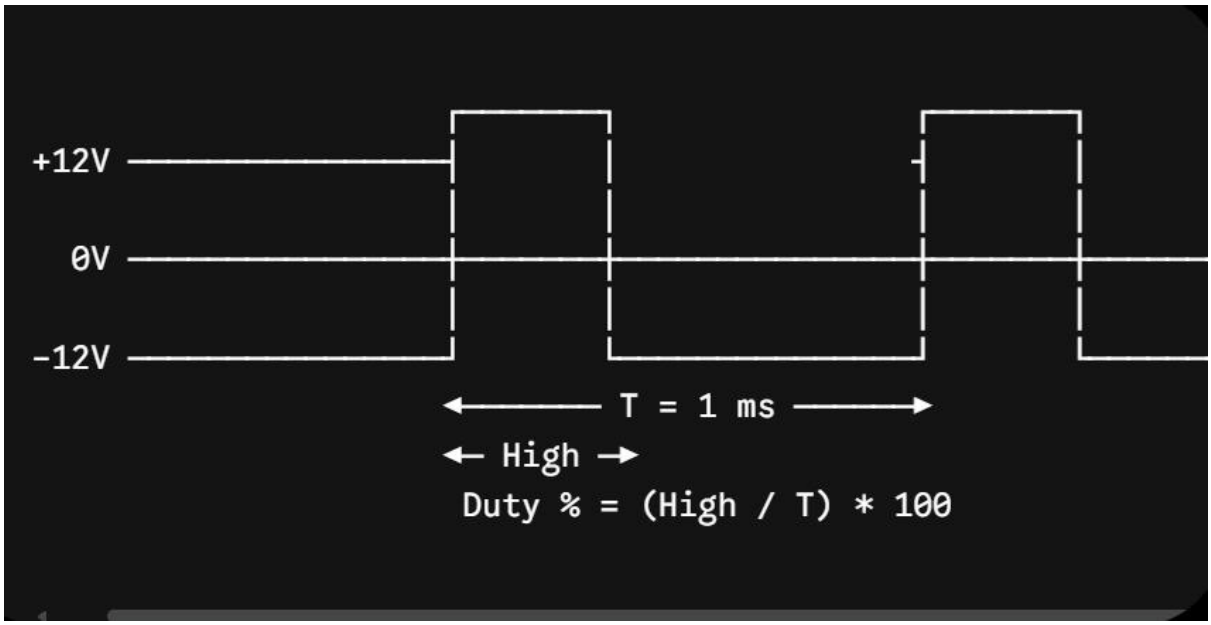
State D----+3v(1khz pwm)---vehicle ready(ventilation required))---evse passes power if outdoor/ventilated, otherwise stays off.



Step 2: setting current liit via pwm duty cycle

Once the ev enters state b or state c , evse generates a 1khz square wave (1000hz) btw +12v and -12v

The duty cycle (percentage of time the signal stays on in each cycle) directly controls the charging current limit(imax)



Standard IEC 61851 Formula Rules

Standard range(10% to 85% duty max current=duty cycle*0.6)

High current range(85% to 96% duty cycle):max current=(duty-64)*2.5

How Dynamic Load Balancing (DLB) Uses CP Signaling

Building load increases: Microcontroller lowers the PWM duty cycle from 53.3% (32 A) down to 16% (10 A).

On-Board Charger (OBC) detects change: Within 5 seconds, the vehicle reads the narrower PWM signal and throttles its power draw down to 10 A.

Building load drops: Microcontroller increases the duty cycle back up, letting the EV draw full speed safely.

Dynamic load balancing: controller calculates p_avail=p_grid,max-p_otherloads

Then distributes according to the selected algorthim

Load management: monitoring and controlling the total electrical load so that the charging station does not exceed its permitted level

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Peak shaving: reducing the mac power drawn from the grod

Without smart charging: 2 kW → 5 kW → 11 kW → 13 kW → 4 kW

With load balancing: 2 kW → 5 kW → 9 kW → 9.5 kW → 4 kW

reduced the peak from 13 kW to 9.5 kW.

Duty Cycle (%)	Allowed Current Limit (Imax)	Common Application
0% to 9%	**0 A**	Charging explicitly prohibited (Pause state).
10%	**6 A**	Minimum standard AC charging limit allowed.
16%	**10 A**	Domestic wall socket safe charging mode.
26.6%	**16 A**	Single-Phase 3.7 kW / Three-Phase 11 kW line.
53.3%	**32 A**	Fast AC Single-Phase 7.4 kW / Three-Phase 22 kW limit.
80%	**48 A**	High-power commercial AC unit limit.
>97%	**Reserved / Fault**	Invalid operating condition.

Load scheduling : deciding when and how much each ev should charge

Charging priority: depends upon state of charge(soc),required departure time, user priority, remaining charging time , user priority,remaning charging time, battery capacity, charging urgency.

State of charge

Soc=remaining battery capacity/max battery capacity *100

25%---battery is relatively empty ; 85%----battery is nearly full

Power allocation: your determines pev1,pev2,pev3…..pevi sub to $\sum Pevi \leq Pstation,max$

Demand response: changing ev charging power in response to grid conditions , electricity prices or system demand.

Energy management system---ems: the higher level controller that decides how the available energy should be distributed

V2G-----vehicle to grid------ev can send power back to the grid as a future scope development

V1G: unidirectional smart charging: the grid charger the ev, but the ev does not send energy back

Software used: matlab and Simulink

Algorithm:

Level1: basic equal power sharing

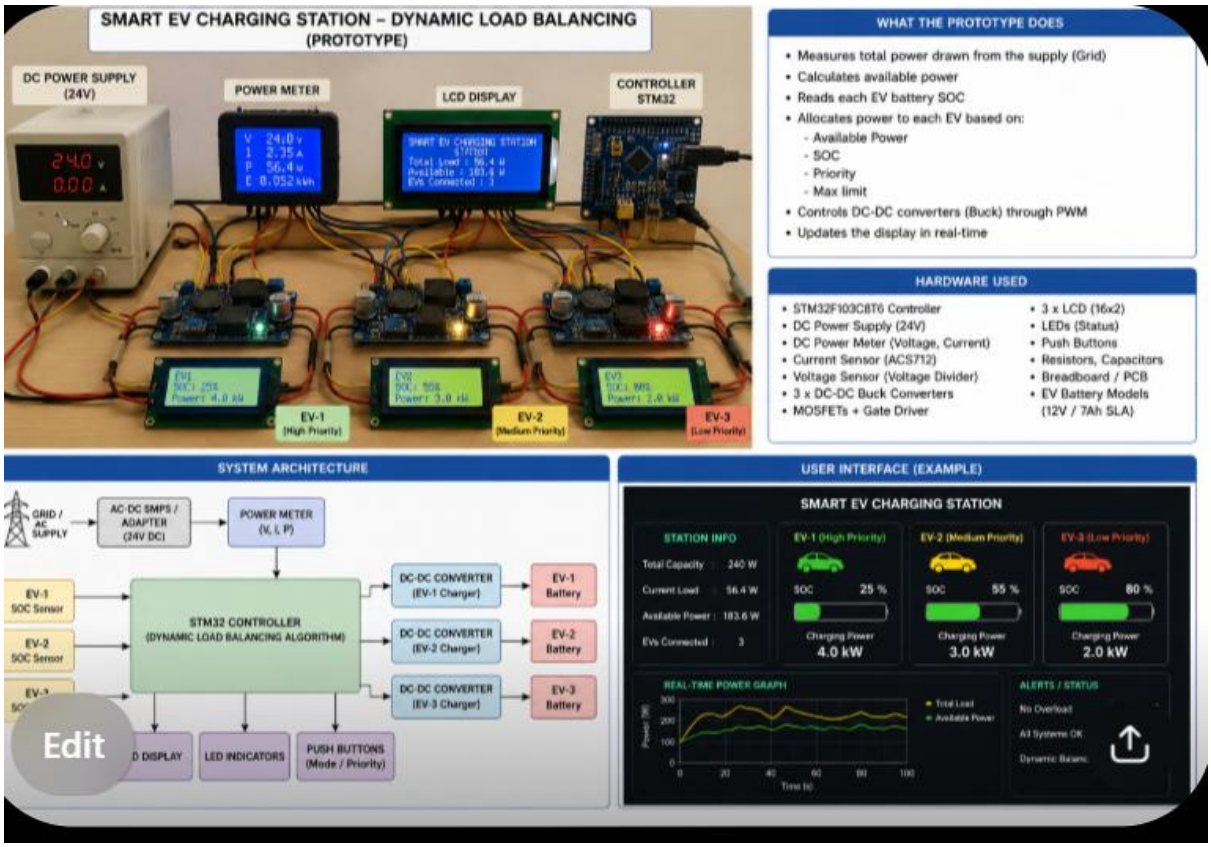
PEV1 =PEV2 =PEV3 =4kW

level2: priority based dynamic load balancing

Priority=w1 (1−SOC)+w2 (Urgency)+w3 (UserPriority)

level3: optimixation based charging

J=w1 (Ppeak)+w2 (Pimbalance)+w3 (Tcharging)+w4 (Cost) and



The research or review paper:

A comprehensive review on electric vechicles smart charging : solutions, vechicles smart charging : solutions, strategies, technologies and challenges. The literature review shows the efficiency of EVSC in reducing charging costs by 30 %, grid operational costs by 10 %, and renewable curtailment by 40 %. The study gives key findings and recommendations which can be helpful for researchers and policymakers.

The economic implication is related to the power exchange between EVs and the power grid through vehicle-to-grid (V2G) technology. The economic implication is related to the power exchange between EVs and the power grid through vehicle-to-grid (V2G) technology

o make EVs genuinely eco-friendly, grids are shifting toward **renewable/green energy sources** (solar, wind, hydro) for EV charging stations.

An EV stores electrical energy in its battery.

For our **prototype**, we'll use a safe low-voltage DC architecture rather than trying to reproduce a real high-power commercial EV charger.

AC charging
AC Grid
↓
EVSE
↓
On-board charger
↓
Battery
The EV's onboard charger performs the AC-to-DC conversion.
DC charging
AC/Grid source
↓
AC/DC
↓
DC Bus
↓
DC/DC
↓
EV Battery

Station
P_available
Each EV
SOCi ,P_request , Pmax
and eventually:
t_departure
Then the controller calculates:
P_allocated

First mathematical constraint

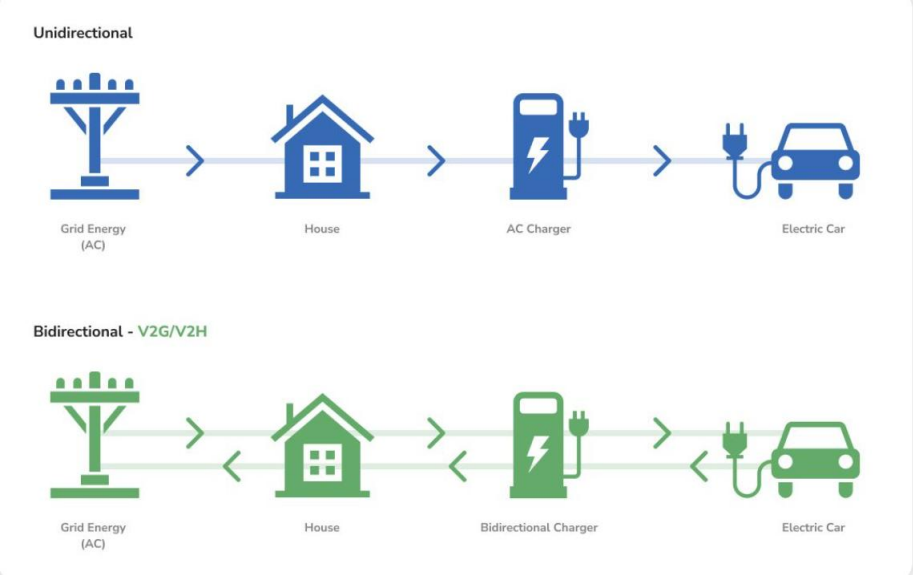
$$\sum_{i=1}^{N} P_{allocated,i} \leq P_{station,max}$$

Eventually our controller can consider

$$Priority_i = f(SOC_i,\ P_{request,i},\ t_{departure,i},\ \ldots)$$

Electrical power p=vi 24*5=120w
Energy E=Pt 120*2=240wh

What Advanced Solutions Can Turn EVs Into Grid Assets?



While smart charging (or V1G) optimizes when EVs draw power, V2G or vehicle-to-grid technology, goes further, allowing EVs to **supply power back to the grid**. In a V2G setup, energy flows both ways, turning EVs into mobile storage units that support the grid during high demand or outages. EVs' large batteries could work like "**mobile power generators**," providing services such as **load shifting, demand response, and peak shaving**. For example, during an evening peak, a utility could signal enrolled EVs to discharge a small portion of their battery back to the grid (for which owners are compensated). This helps with grid relief.

A California Energy Commission official stated, _"Through vehicle-to-grid integration, electric vehicles represent substantially more potential for grid benefits than any other distributed resource."_ That's a bold claim, underscoring how a fleet of EVs, in aggregate, could exceed even stationary batteries or solar in providing flexible grid support, simply due to the scale of the automotive market.

the **technology behind V2G** involves bidirectional charging hardware and smart software. Communication protocols like ISO 15118-20 coordinate energy dispatch. Several countries have pilot programs proving the concept.

In Europe, especially the UK, V2G is gaining traction. The UK hosted **over half of global V2G projects** as of 2023 (131 projects, 6,800 bidirectional charge points) thanks to supportive policies and high EV penetration

Japan pioneered V2G with the Nissan Leaf, and China is exploring V2G to manage projected peak demand (**600 GW peak from EVs by 2030**). The US too has V2G trials with utilities like PG&E and research by NREL, focusing on services like frequency regulation and peak shaving.

Power
How quickly energy is being transferred.
Example:
Energy
How much energy has actually been transferred.
kW = rate
kWh = amount

Which information should our STM32 consider before deciding how to reduce the charging powers?

Ah = Ampere-hour, and it represents the **amount of electric charge the battery can store/deliver**
For a **40 Ah** battery:

- If it delivers **1 A** continuously → theoretically about **40 hours**
- If it delivers **10 A** → theoretically about **4 hours**
- If it delivers **20 A** → theoretically about **2 hours**

: **Ah tells us how much charge is involved, while A tells us the instantaneous current.**

Suppose our battery is:

$$48V, \quad 40Ah$$

Approximate stored energy:

$$E = V \times Ah$$

$$E = 48 \times 40$$

$$\boxed{E \approx 1920Wh = 1.92kWh}$$

Now suppose our charger supplies:

$$I = 10A$$

Charging power is:

$$P = VI$$

$$P = 48 \times 10$$

$$\boxed{P = 480W}$$

So:

40 Ah → battery charge capacity

10 A → charging current at that moment

480 W → charging power at that moment

one important EV detail: **real batteries don't usually charge at a constant 10 A for the entire charging cycle.** Charging current typically changes as the battery approaches full charge.

SOC = State of Charge

It tells us **how full the battery currently is**, usually expressed as a percentage.

SOC	Meaning
0%	Empty
25%	Low
50%	Half charged
80%	Mostly charged
100%	Fully charged

For our DLB system, SOC is extremely important.

EV	SOC
EV1	20%
EV2	60%
EV3	90%

If all three request 300 W, should we necessarily give them equal power?

Not necessarily.

EV1 may deserve higher priority because its SOC is much lower.

That's where our project becomes intelligent.

Basic SOC equation

For a simple battery model:

$$SOC(t) = SOC(t_0) + \frac{1}{C_{Ah}} \int_{t_0}^{t} I(\tau) d\tau$$

For charging, SOC increases.

For discharging, SOC decreases.

For our first MATLAB model, we'll use a simplified discrete equation:

$$SOC_{new} = SOC_{old} + \frac{I \Delta t}{C_{Ah}}$$

I = charging current

t = time interval

Cah = battery capacity

And that's exactly why the four things you identified earlier—

$$SOC,\ urgency,\ departure\ time,\ power\ required$$

—are going to become inputs to our **dynamic power allocation algorithm**.

C-rate connects battery capacity to charging current.

C-rate tells us how fast a battery is being charged or discharged relative to its capacity.

The basic relationship is:

$$I = C\text{-rate} \times C_{Ah}$$

where C_{Ah} is the battery capacity.

C-rate Current Approx. ideal full-charge time

0.25C 10 A 4 h

0.5C 20 A 2 h

1C 40 A 1 h

2C 80 A 0.5 h

These are idealized calculations, not a promise that a real EV battery can safely charge at those rates. Real charging depends on the battery chemistry, BMS, temperature, SOC, charger and manufacturer's limits.

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4. Why C-rate matters for OUR project

Suppose our station has: Pmax =10kW

If our controller says:

"EV1 needs more power, so I'll keep increasing current.".

The battery has charging limits.

So our controller eventually needs constraints

$$I_i \leq I_{max,i}$$

and therefore:

$$P_i \leq P_{max,i}$$

Our dynamic load balancing problem is actually becoming:

$$\text{Maximize useful charging}$$

subject to:

$$\sum P_i \leq P_{station,max}$$

and:

$$P_i \leq P_{EV,i,max}$$

and eventually:

$$SOC_i \leq SOC_{target,i}$$

So increasing C-rate means increasing charging current and therefore charging power, assuming voltage is approximately fixed.

$$P = VI$$

ppose our battery is:

$$48V, \quad 40Ah$$

0.25C:

$$I = 10A$$

herefore:

$$P = 48 \times 10$$

$$\boxed{P = 480W}$$

0.5C:

$$I = 20A$$

:

$$P = 48 \times 20$$